TREEWALK NYC • STEAM LESSON PLAN

Littleleaf Linden

Tilia cordata

Species background: Native to Europe and western Asia, long planted along grand boulevards -- Berlin's famous street 'Unter den Linden' is named after this tree. The fruit hangs from a pale, papery bract that acts like a wing, letting the wind spin the whole cluster away from the parent tree -- look up in late summer to spot them.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

Littleleaf Linden: Identify littleleaf linden by its lopsided heart-shaped leaf and sweet-smelling flowers.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, crayons.

Teacher background

Find a leaf shaped like a lopsided heart. Linden flowers smell sweet and draw bees on warm days -- follow your nose before you find the tree.

Procedure

1 Engage: Ask what a heart shape looks like, then find the 'lopsided' part on a real or pictured leaf.
2 Explore: If it's summer, smell the air near the tree before looking up.
3 Explain: Explain that bees love the flower's sweet smell just as much as we do.
4 Art: Trace a heart-shaped leaf and draw a small bee next to it.
5 Evaluate: Student can describe the leaf as a lopsided heart and name one animal drawn to the flowers.

Discussion questions

What other smells help you find things without looking?

Assessment

Correctly describes the leaf shape and names bees as visitors.

Extension

Look for bees near any flowering tree on your walk.

GRADE BAND 3-5

Littleleaf Linden: Describe the linden's two different dispersal and pollination strategies.

Standards connection: NGSS 3-LS4-2; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, colored pencils.

Teacher background

A pale bract 'wing' carries the fruit on wind like a tiny helicopter, while the flowers below rely on scent and nectar to recruit pollinators -- two different strategies from one tree.

Procedure

1. Engage: Ask how a seed could travel without an animal carrying it.
2. Explore: Look for the pale, papery wing-shaped bract in late summer (real or pictured).
3. Explain: Contrast wind dispersal (the wing) with pollinator attraction (the scent), as two separate jobs the tree solves differently.
4. Art: Draw the lopsided heart-shaped leaf, then write one line inside it about something you love about trees.
5. Evaluate: Student names the two strategies (wind dispersal, scent pollination) and which part of the tree does each job.

Discussion questions

Why might a tree need two completely different strategies instead of just one?

Assessment

Correctly names both strategies and matches each to the right plant structure.

Extension

Find another tree's seed on your walk and guess how it travels -- wind, animal, or water.

GRADE BAND 6-8

Littleleaf Linden: Explain nectar reward economics and this tree's role in supporting a single-source honey.

Standards connection: NGSS MS-LS2-5; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, notebook.

Teacher background

Linden's flowers are one of the most important late-season nectar sources for city bees, and linden honey is prized for its distinct minty-floral flavor -- productive enough to be sold as its own honey variety.

Procedure

1. Engage: What would it take for a honey to be sold as a single 'variety' instead of a blend?
2. Explore: Research how much nectar a linden tree can produce during its bloom period.
3. Explain: Connect nectar volume and reliability to why beekeepers place hives near linden groves.
4. Art: Write a short poem or song about the smell of linden flowers in summer, then illustrate it with the heart-shaped leaf as a border.
5. Evaluate: Write a paragraph explaining why this tree can support a single-source honey when many trees can't.

Discussion questions

What does the existence of 'linden honey' as a distinct product tell you about this tree's ecological role?

Assessment

Paragraph correctly connects nectar volume/reliability to single-source honey production.

Extension

Research one other single-source honey (clover, orange blossom, buckwheat) and compare the plant behind it.

GRADE BAND 9-12

Littleleaf Linden: Evaluate competing hypotheses for reported bee die-offs beneath heavily blooming lindens using primary sources.

Standards connection: NGSS HS-LS2-7; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, access to research sources, fine-liner pens.

Teacher background

Linden produces enough nectar to support single-source 'linden honey.' Some cities have documented reports of bee die-offs beneath heavily blooming lindens, and researchers have proposed competing hypotheses -- nectar toxicity versus simple overexertion -- to explain them.

Procedure

1. Engage: When a scientific question has multiple competing hypotheses, how should a careful researcher evaluate them?
2. Explore: Research the documented linden bee die-off reports and identify the evidence behind each competing hypothesis.
3. Explain: Discuss why 'nectar toxicity' and 'simple overexertion' would produce similar-looking evidence (dead bees under the tree) but require different kinds of proof.
4. Art: Design a scent-inspired abstract color study representing the smell of linden blossoms without drawing the flower literally, then explain your color choices in one paragraph.
5. Evaluate: Submit a short position paper weighing the two hypotheses using specific cited evidence.

Discussion questions

What experiment could help distinguish between the toxicity hypothesis and the overexertion hypothesis?

Assessment

Position paper evaluated on reasoning quality and use of specific evidence, not on which hypothesis is favored.

Extension

Research how urban beekeepers manage hive placement relative to linden bloom timing in light of this open question.

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.